METRIC
A-A-52163B
December 4, 2012
SUPERSEDING
A-A-52163A
May 29, 1996

# COMMERCIAL ITEM DESCRIPTION

# LADDERS, VEHICLE-BOARDING

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This CID covers nine different types of portable, metal, step-ladders (hereinafter referred to as ladders) for use by personnel in boarding vehicles.

2. SALIENT CHARACTERISTICS

2.1 Materials. The materials used for the ladders shall be either aluminum or steel as specified. However, the specific material compositions of the aluminum and steel shall be in accordance with the acceptable commercial material specifications for metal ladders. Component corrosion protection shall be equal to or exceed that provided by hot dip galvanized 1020 steel with coating thickness in accordance with ASTM A123/A123M.

2.2 Design and construction. The ladders shall conform to the physical and dimensional requirements specified in figures 1 thru 9. The ladders shall also conform to the applicable requirements of ANSI A14.2 for portable metal ladders. When applicable, drawings and dimensions of ladder components that are needed, but not shown in figures 1 thru 9, shall be provided by the procuring activity (see 6.2). The ladders shall conform to the applicable requirements of ANSI A14.2.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or sent by email to usarmy.detroit.rdecom.mbx.tardec-standardization@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2.2 Dimensional tolerances. Unless otherwise specified, the dimensions specified in figures 1 thru 9 shall have the following tolerances:

a. For dimensions less than or equal to 10 mm, tolerances shall be ±0.4 mm and ±1 degree (°).
b. For dimensions greater than 10 mm and less than or equal to 100 mm, tolerances shall be ±0.8 mm and ±1°.
c. For dimensions greater than 100 mm, tolerances shall be ±2 mm and ±1°.

2.3 Weight. The total weight of any ladder shall not exceed 18 kilograms (kg) [40 pounds (lb)].

2.4 Load. Each step of every ladder shall be capable of withstanding a 360 kg (800 lb) load without damage or permanent set to any of the ladder parts.

2.5 Welding. Welding procedures shall be performed in accordance with the manufacturer's standards for welds. In no event shall processes such as peening and plugging be used for reclaiming any part of the ladder without authorization from the procuring activity. Warpage shall be limited to 6.4 millimeters (mm) [0.25 inches (in)] out of plane. There shall be no lack of fusion which causes failure.

2.5.1 Welded surfaces. All welded parts shall be clean and free of dirt, sand, scale, and other extraneous materials. The welded surfaces shall be free of burrs, cracks, and inclusions. All sharp corners shall be rounded or beveled, except for those on the non-skid step surfaces.

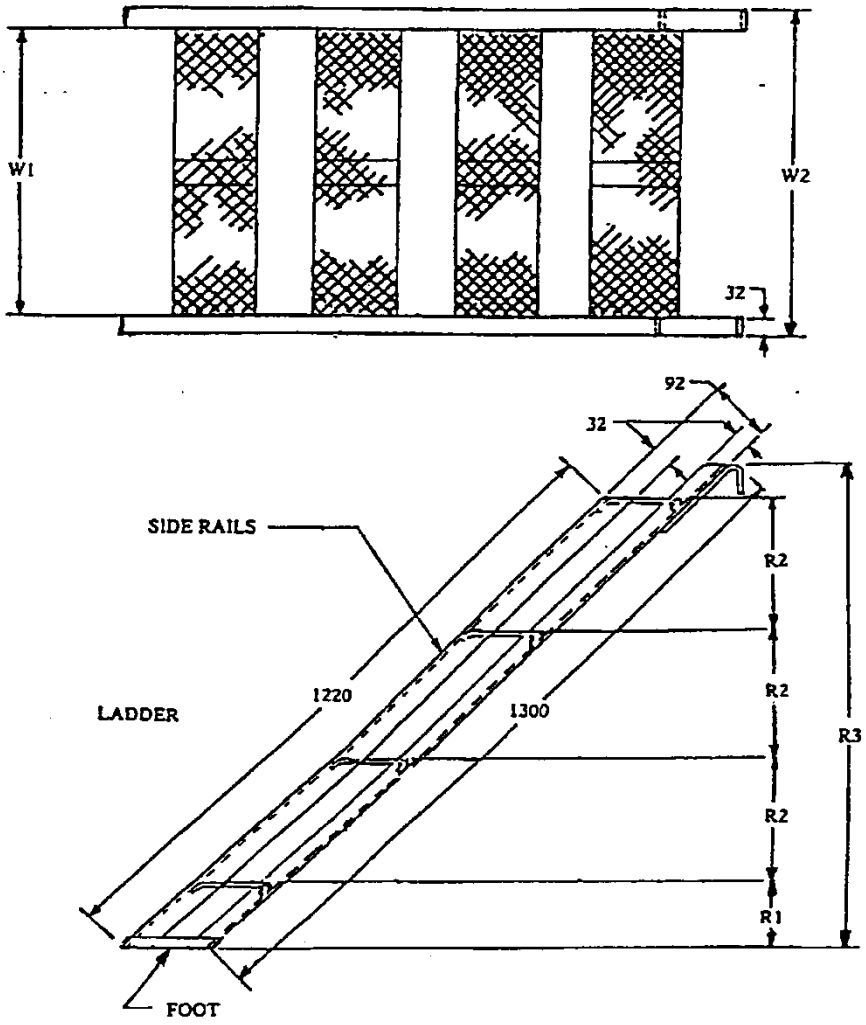
2.6 Non-skid surfaces. All steps shall be constructed of material which has been formed in a manner to provide for a permanent non-skid surface.

2.7 Coatings. The ladders shall be cleaned, pretreated, primed, and painted in accordance with the accepted commercial standard practice. Unless otherwise specified (see 6.2), the base topcoat color shall be green 383, chip number 34094 of FED-STD-595. If CARC paint or camouflage patterns are required, it will be specified in the contract or order (see 6.2).

2.8 Warning labels. Ladders shall have warning labels that contain "Danger" markings in accordance with appendix A of ANSI 14.2, warning personnel that the ladder will conduct electricity. The label design and color shall conform to this standard.



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| | | |
|---|---|---|
| PIN number | | A52163-A1 |
| Number of steps | | 4 |
| Width, inside rails | W1 | 479 mm (18.9 in) |
| Width, outside rails | W2 | 543 mm (21.4 in) |
| First step rise | R1 | 102 mm (4 in) |
| Typical step rise | R2 | 254 mm (10 in) |
| Vertical rise, total | R3 | 921 mm (36.2 in) |
| Angle with ground | A | 45 degrees (°) |
| Material | | Steel |

FIGURE 1. Ladder assembly, type A.



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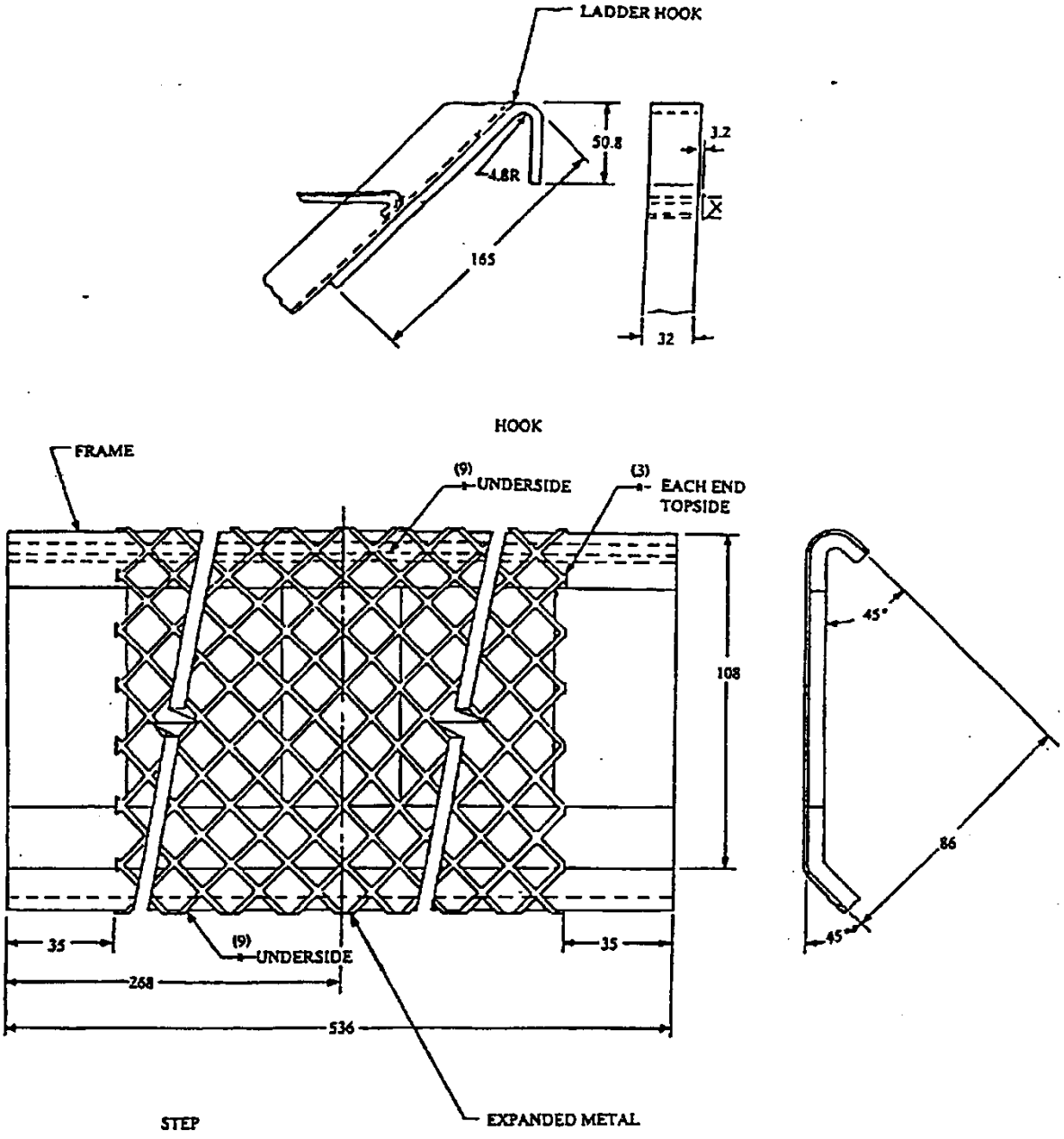
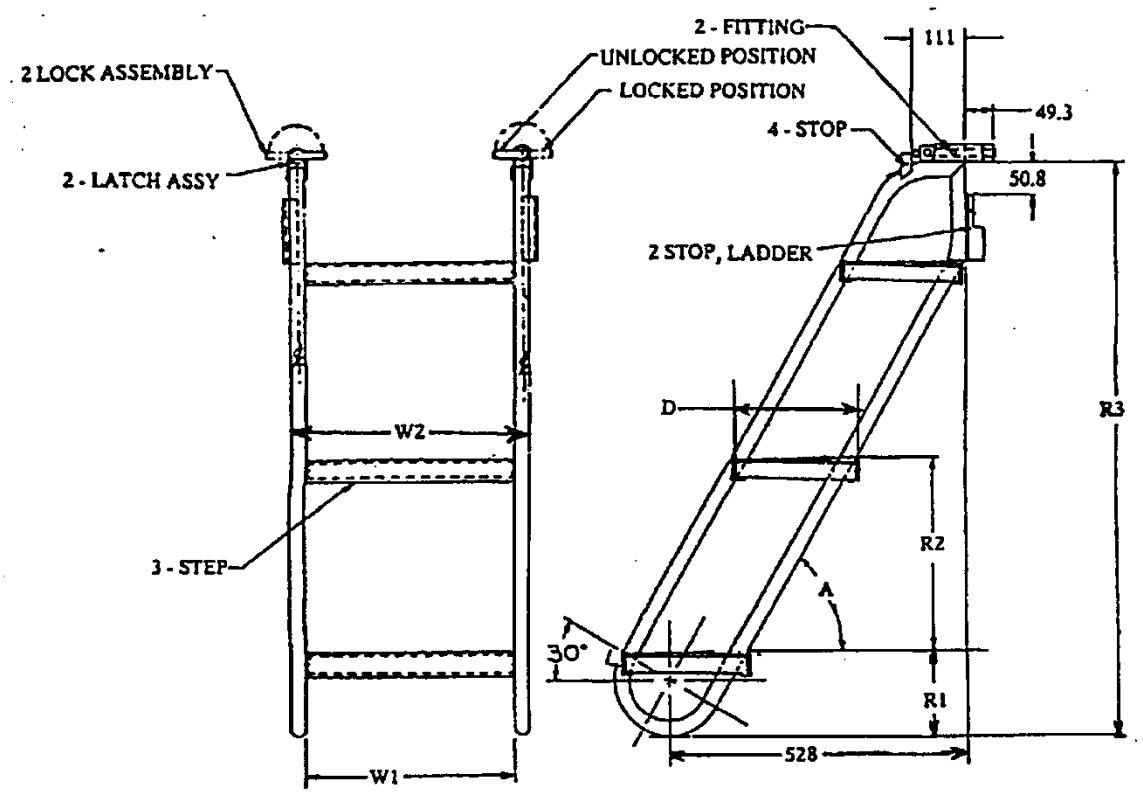



FIGURE 1. Ladder assembly, type A - Continued.



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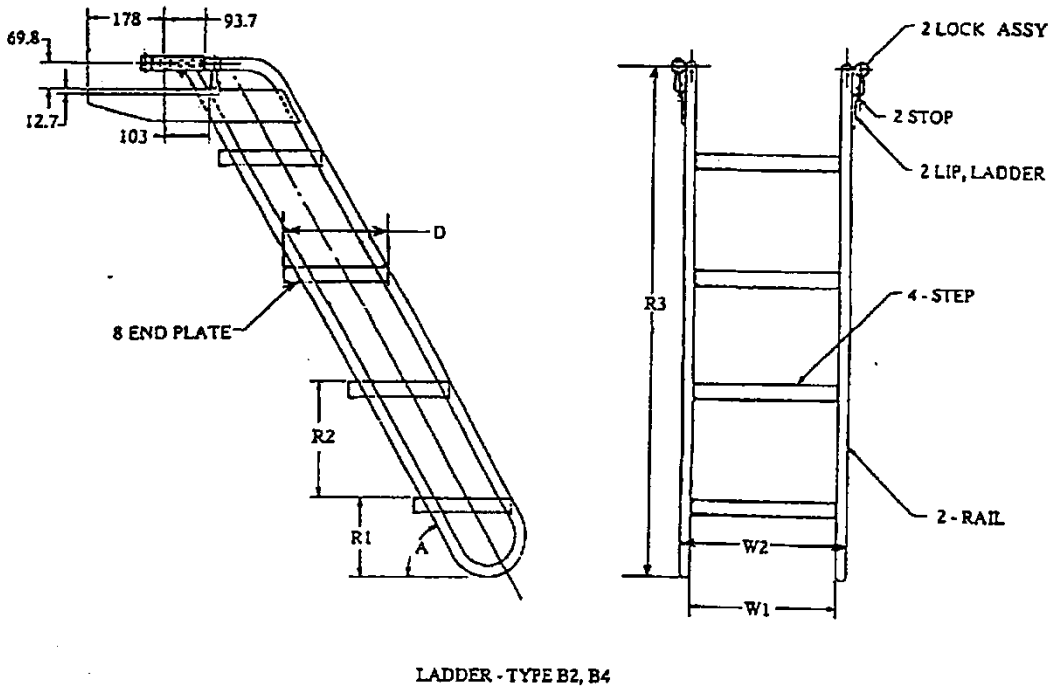

LADDER - TYPE B1, B3

| PIN number | | A52163-B1 | A52163-B2 | A52163-B3 | A52163-B4 |
|---|---|---|---|---|---|
| Number of steps | | 3 | 4 | 3 | 4 |
| Width, inside rails | W1 | 508 mm (20 in) | 508 mm (20 in) | 359 mm (14.1 in) | 359 mm (14.1 in) |
| Width, outside rails | W2 | 559 mm (22 in) | 559 mm (22 in) | 410 mm (16.1 in) | 410 mm (16.1 in) |
| Step depth | D | 241 mm (9.5 in) | 241 mm (9.5 in) | 241 mm (9.5 in) | 241 mm (9.5 in) |
| First step rise | R1 | 146 mm (5.8 in) | 184 mm (7.25 in) | 146 mm (5.8 in) | 184 mm (7.25 in) |
| Typical step rise | R2 | 330 mm (13 in) | 267 mm (10.5 in) | 330 mm (13 in) | 267 mm (10.5 in) |
| Vertical rise, total | R3 | 987 mm (38.9 in) | 1190 mm (46.8 in) | 987 mm (38.9 in) | 1190 mm (46.8 in) |
| Angle with ground | A | 60° | 60° | 60° | 60° |
| Material, rail | | Steel tubing | Steel tubing | Steel tubing | Steel tubing |
| Material, step | | Steel | Steel | Steel | Steel |

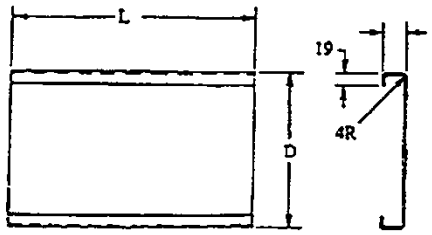
FIGURE 2. Ladder assembly, type B.



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LADDER - TYPE B2, B4



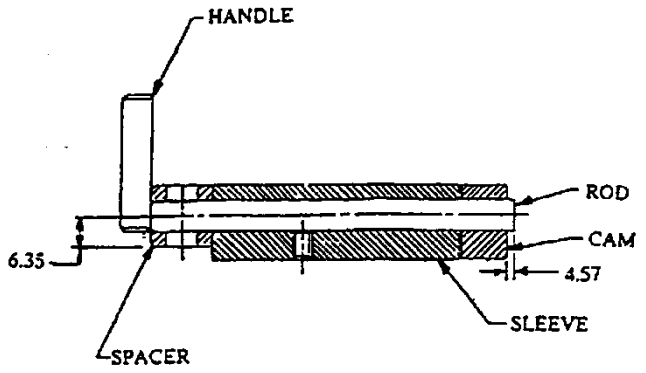

| PIN NUMBER | L |
|---|---|
| A52163-B1, B2 | 508 |
| A52163-B3, B4 | 356 |

STEP

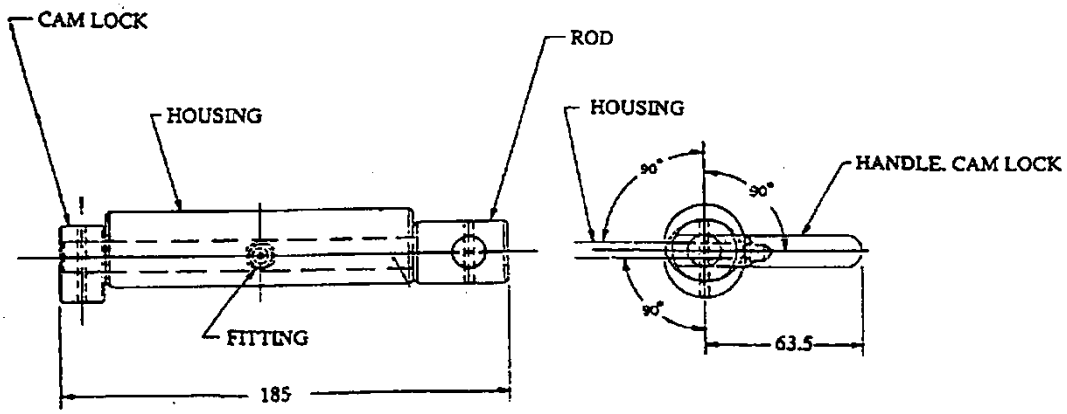
FIGURE 2. Ladder assembly, type B - Continued.



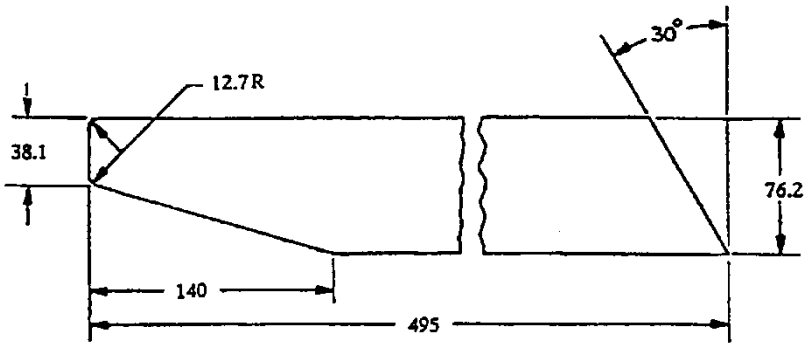
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LATCH ASSY - TYPE B1, B3




LOCK ASSEMBLY -- TYPE B2, B4

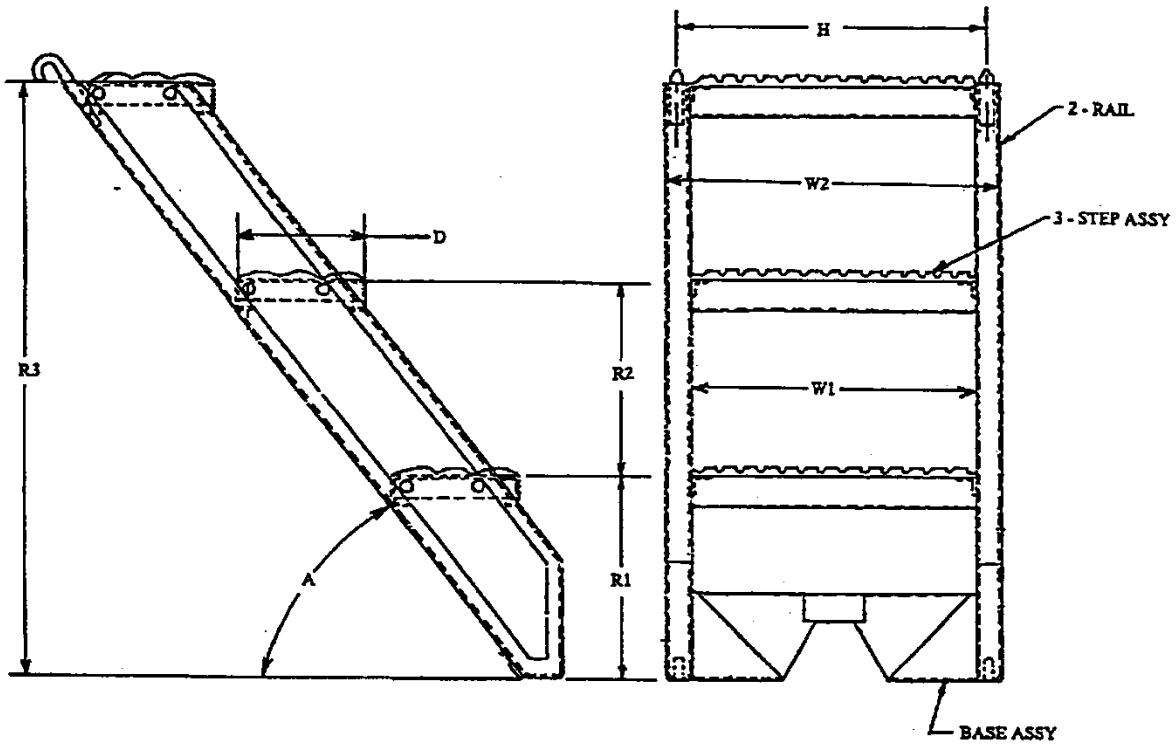



LIP - TYPE B2, B4

FIGURE 2. Ladder assembly, type B - Continued.



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| | | |
|---|---|---|
| PIN number | | A52163-C1 |
| Number of steps | | 3 |
| Width, inside rails | W1 | 400 mm (15.8 in) |
| Width, outside rails | W2 | 470 mm (18.5 in) |
| Hook centers | H | 435 mm (17.1 in) |
| Step depth | D | 184 mm (7.25 in) |
| First step rise | R1 | 293 mm (11.5 in) |
| Typical step rise | R2 | 280 mm (11 in) |
| Vertical rise, total | R3 | 853 mm (33.6 in) |
| Angle with ground | A | 52° |
| Material | | Aluminum alloy |

FIGURE 3. Ladder assembly, type C.



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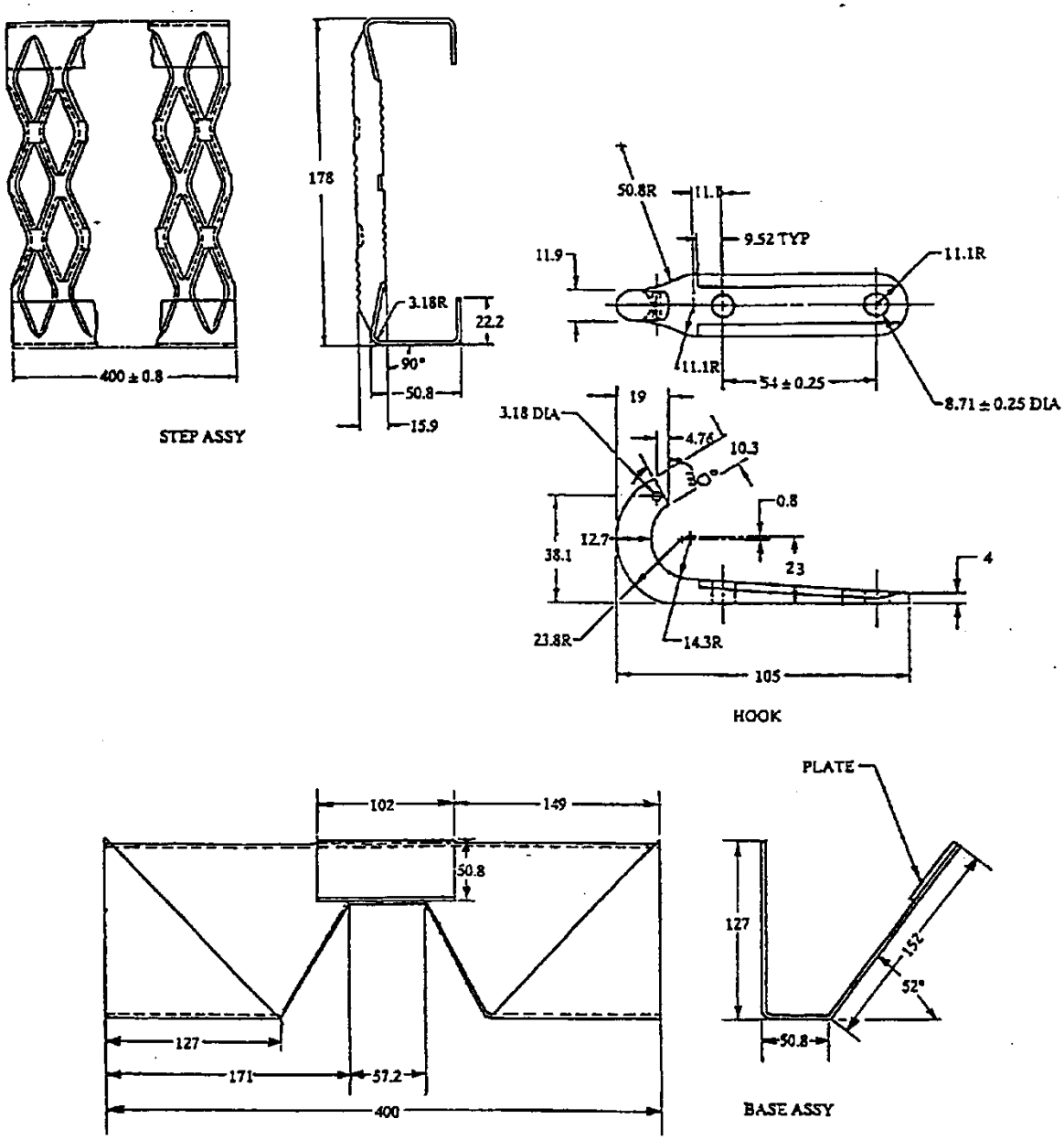
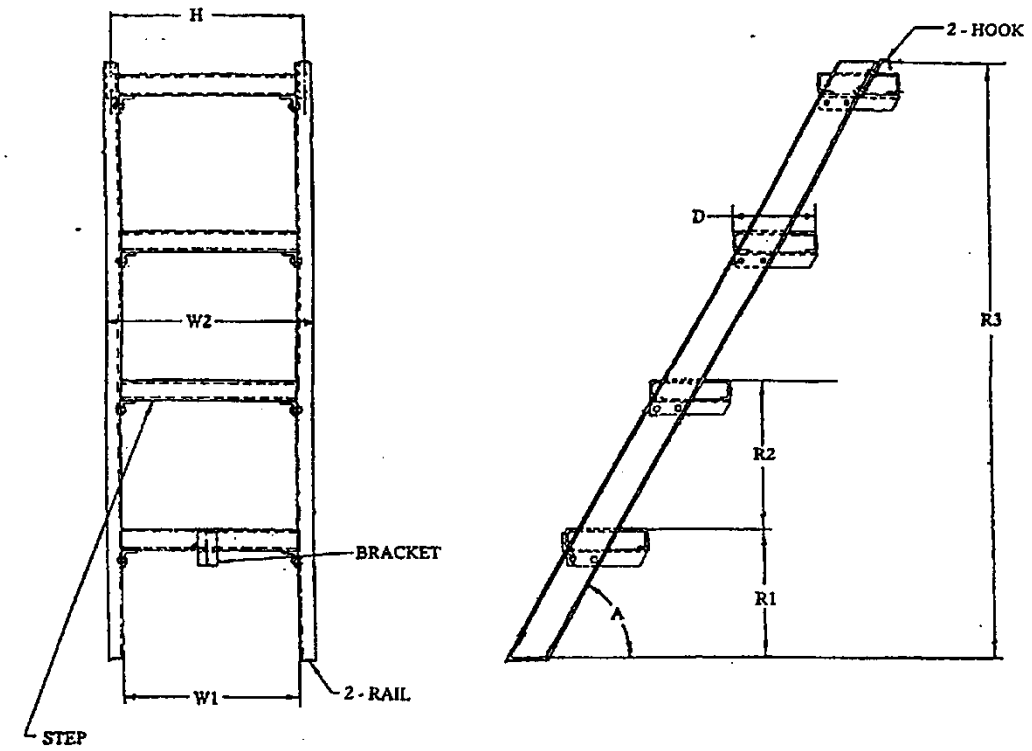



FIGURE 3. Ladder assembly, type C - Continued.



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| PIN number | | A52163-D1 | A52163-D2 | A52163-D3 | A52163-D4 |
|---|---|---|---|---|---|
| Number of steps | | 4 | 5 | 4 | 4 |
| Width, inside rails | W1 | 457 mm (18 in) | 387 mm (15.2 in) | 508 mm (20 in) | 400 mm (15.8 in) |
| Width, outside rails | W2 | 537 mm (21.1 in) | 457 mm (18 in) | 610 mm (24 in) | 464 mm (18.2 in) |
| Hook centers | H | 502 mm (19.8 in) | 432 mm (17 in) | 565 mm (22.2 in) | 432 mm (17 in) |
| Step depth | D | 203 mm (8 in) | 159 mm (6.25 in) | 178 mm (7 in) | 178 mm (7 in) |
| First step rise | R1 | 286 mm (11.2 in) | 129 mm (5.08 in) | 254 mm (10 in) | 270 mm (10.6 in) |
| Typical step rise | R2 | 280 mm (11 in) | 254 mm (10 in) | 254 mm (10 in) | 280 mm (11 in) |
| Vertical rise, total | R3 | 1220 mm (48 in) | 1150 mm (45.1 in) | 1060 mm (41.8 in) | 1300 mm (51.3 in) |
| Angle with ground | A | 60° | 53° | 45° | 60° |
| Material | | Steel | Steel | Aluminum | Aluminum |

FIGURE 4. Ladder assembly, type D.



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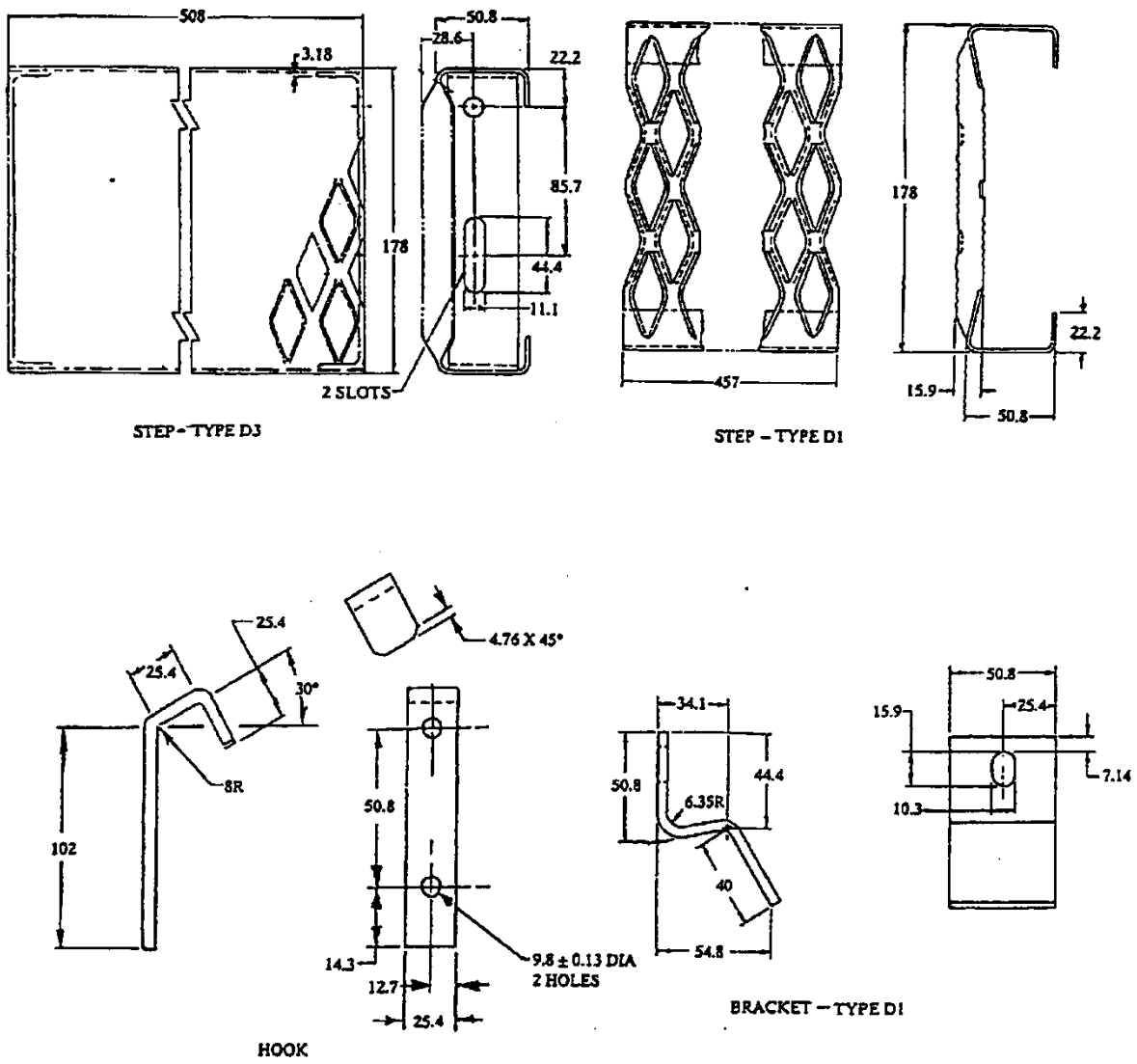
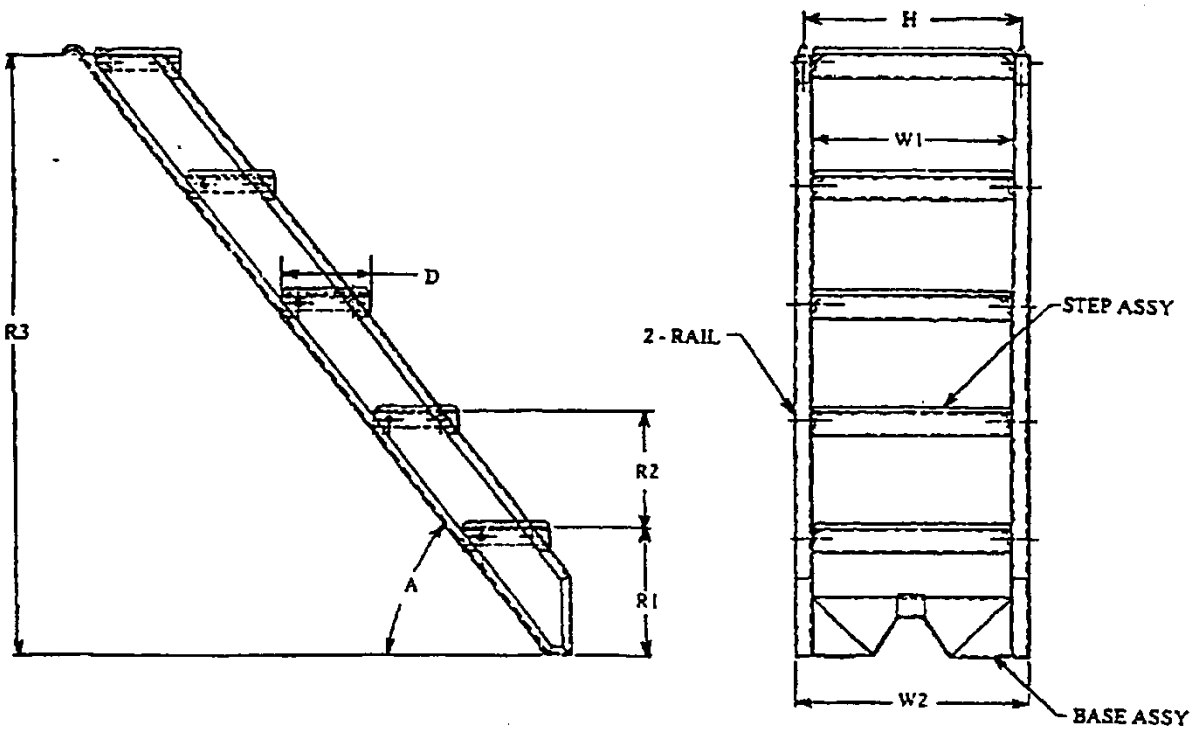



FIGURE 4. Ladder assembly, type D - Continued.



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| PIN number | | A52163-E1 | A52163-E2 |
|---|---|---|---|
| Number of steps | | 5 | 5 |
| Width, inside rails | W1 | 457 mm (18 in) | 400 mm (15.8 in) |
| Width, outside rails | W2 | 527 mm (20.8 in) | 470 mm (18.5 in) |
| Hook centers | H | 492 mm (19.4 in) | 435 mm (17.1 in) |
| Step depth | D | 178 mm (7 in) | 178 mm (7 in) |
| First step rise | R1 | 240 mm (9.54 in) | 240 mm (9.45 in) |
| Typical step rise | R2 | 254 mm (10 in) | 254 mm (10 in) |
| Vertical step rise | R3 | 1256 mm (49.4 in) | 1256 mm (49.4 in) |
| Angle with ground | A | 53° | 53° |
| Material | | Aluminum | Aluminum |

FIGURE 5. Ladder assembly, type E.



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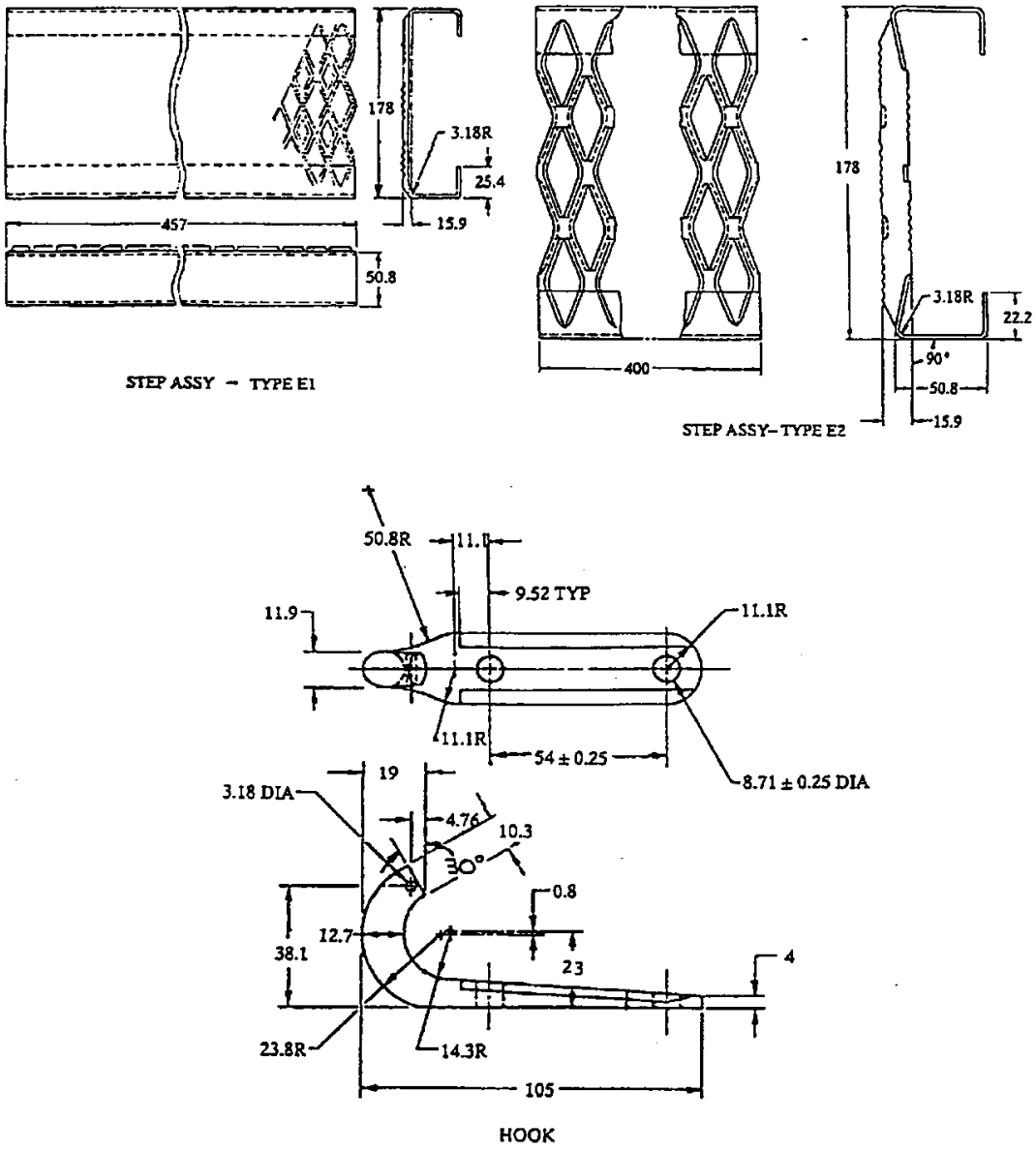



FIGURE 5. Ladder assembly, type E - Continued.



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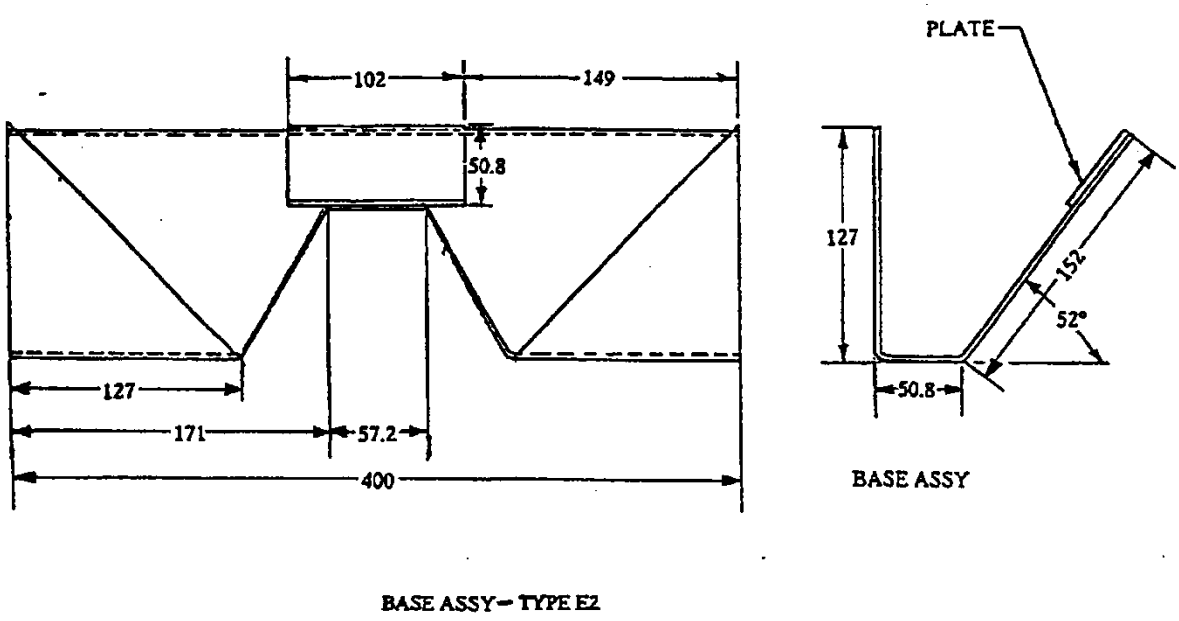
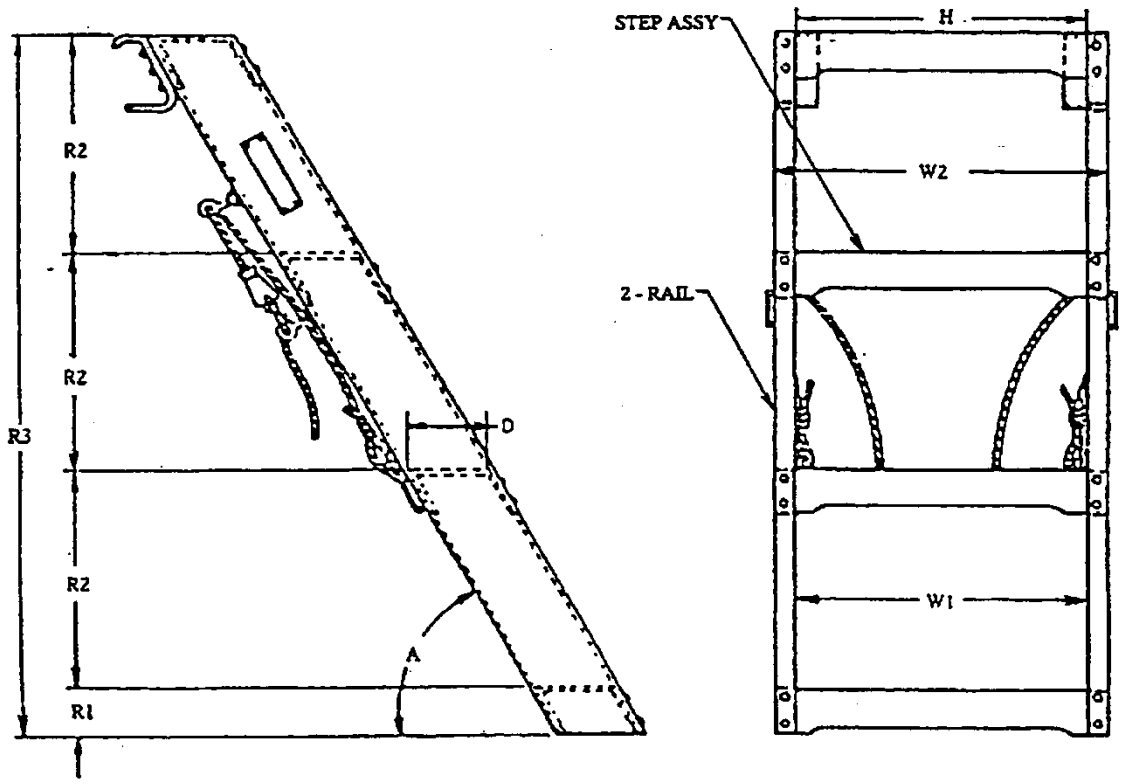



FIGURE 5. Ladder assembly, type E -Continued.



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| | | | |
|---|---|---|---|
| PIN number | | A52163-F1 | A52163-F2 |
| Number of steps | | 4 | 6 |
| Width, inside rails | W1 | 365 mm (14.4 in) | 365 mm (14.4 in) |
| Width, outside rails | W2 | 416 mm (16.4 in) | 416 mm (16.4 in) |
| Hook centers | H | 365 mm (14.4 in) | 365 mm (14.4 in) |
| Step depth | D | 109 mm (4.28 in) | 109 mm (4.28 in) |
| First step rise | R1 | 46.8 mm (1.84 in) | 46.8 mm (1.84 in) |
| Typical step rise | R2 | 280 mm (11 in) | 280 mm (11 in) |
| Vertical rise, total | R3 | 883 mm (34.8 in) | 1370 mm (54 in) |
| Angle with ground | A | 60° | 60° |
| Material | | Aluminum | Aluminum |

FIGURE 6. Ladder assembly, type F.



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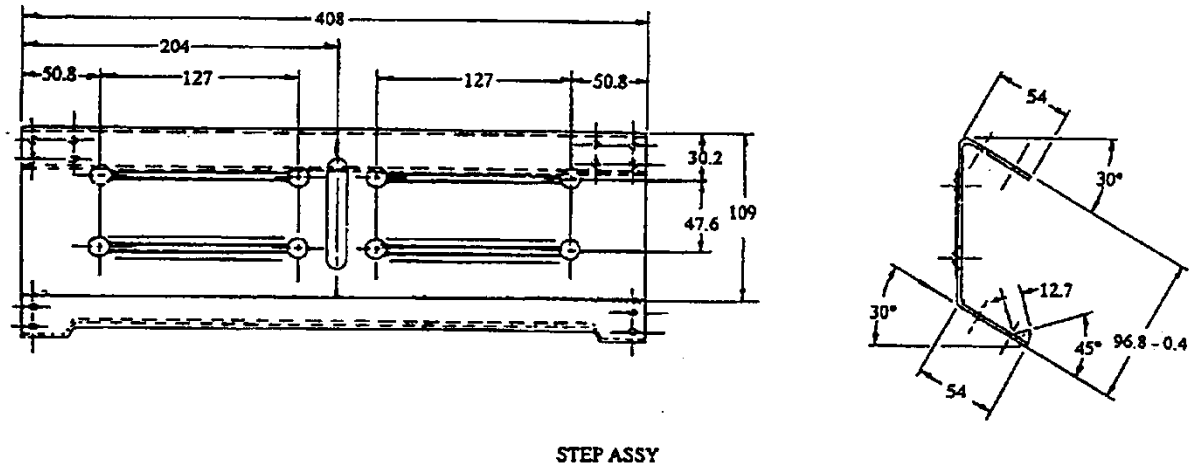
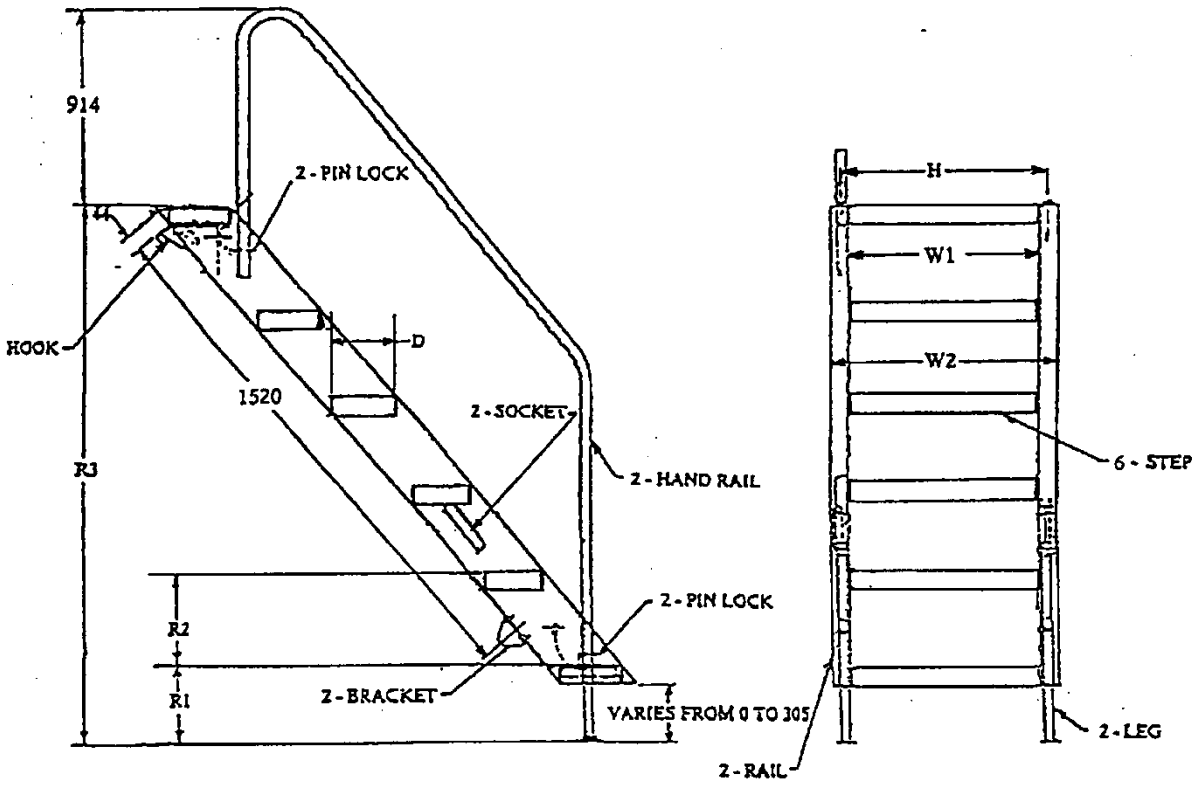



FIGURE 6. Ladder assembly, type F - Continued.



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| | | |
|---|---|---|
| PIN number | | A52163-G1 |
| Number of steps | | 6 |
| Width, inside rails | W1 | 508 mm (20 in) |
| Width, outside rails | W2 | 610 mm (24 in) |
| Hook centers | H | 560 mm (22 in) |
| Step depth | D | 180 mm (7 in) |
| First step rise (adjustable) | R1 | 57.2 to 362 mm (2.25 to 14.2 in) |
| Typical step rise | R2 | 274 mm (10.8 in) |
| Vertical rise, total | R3 | 1370 to 1675 mm (53.9 to 66 in) |
| Angle with ground | A | 51° |
| Material | | Aluminum |

FIGURE 7. Ladder assembly, type G.



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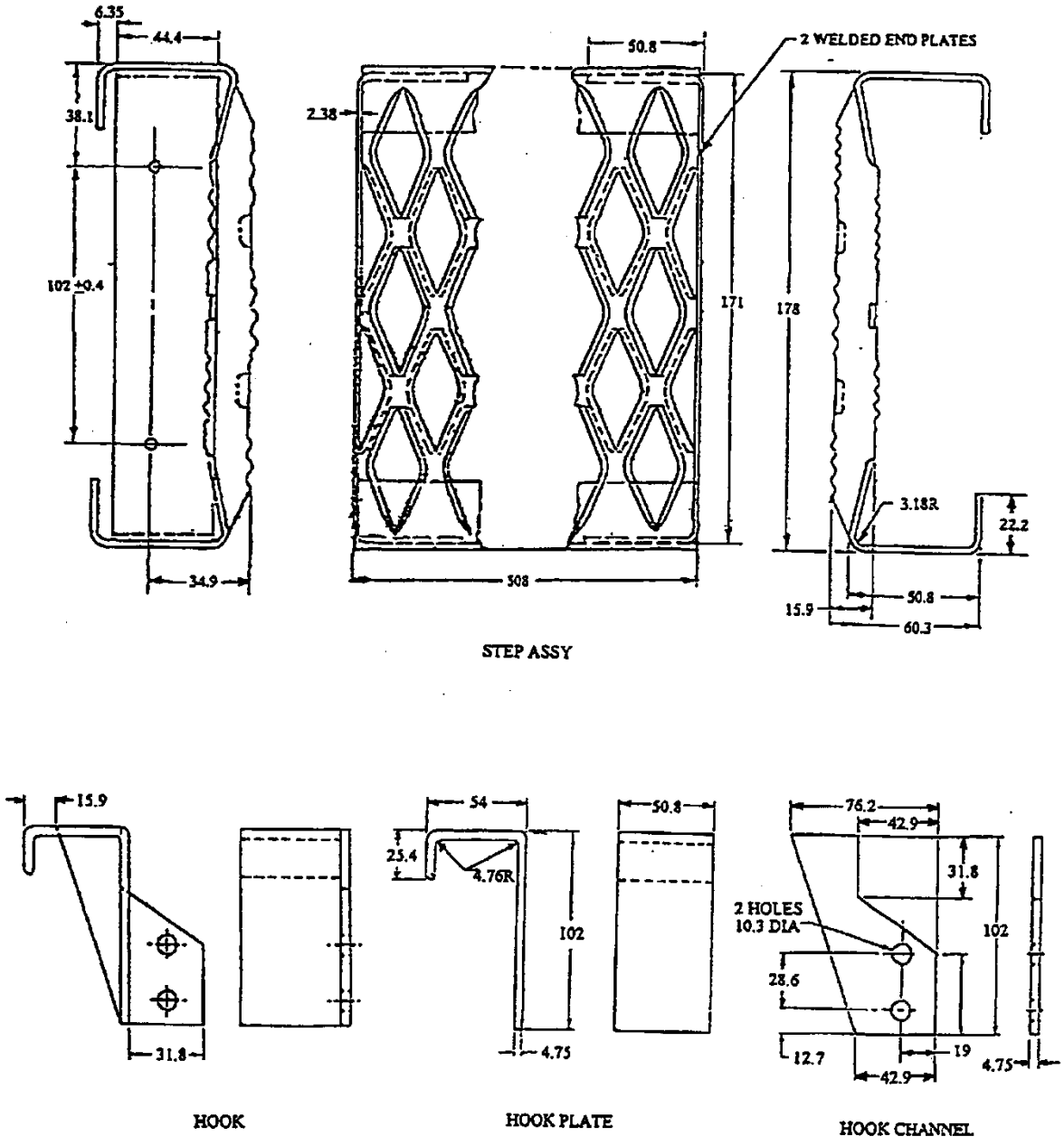



FIGURE 7. Ladder assembly, type G - Continued.



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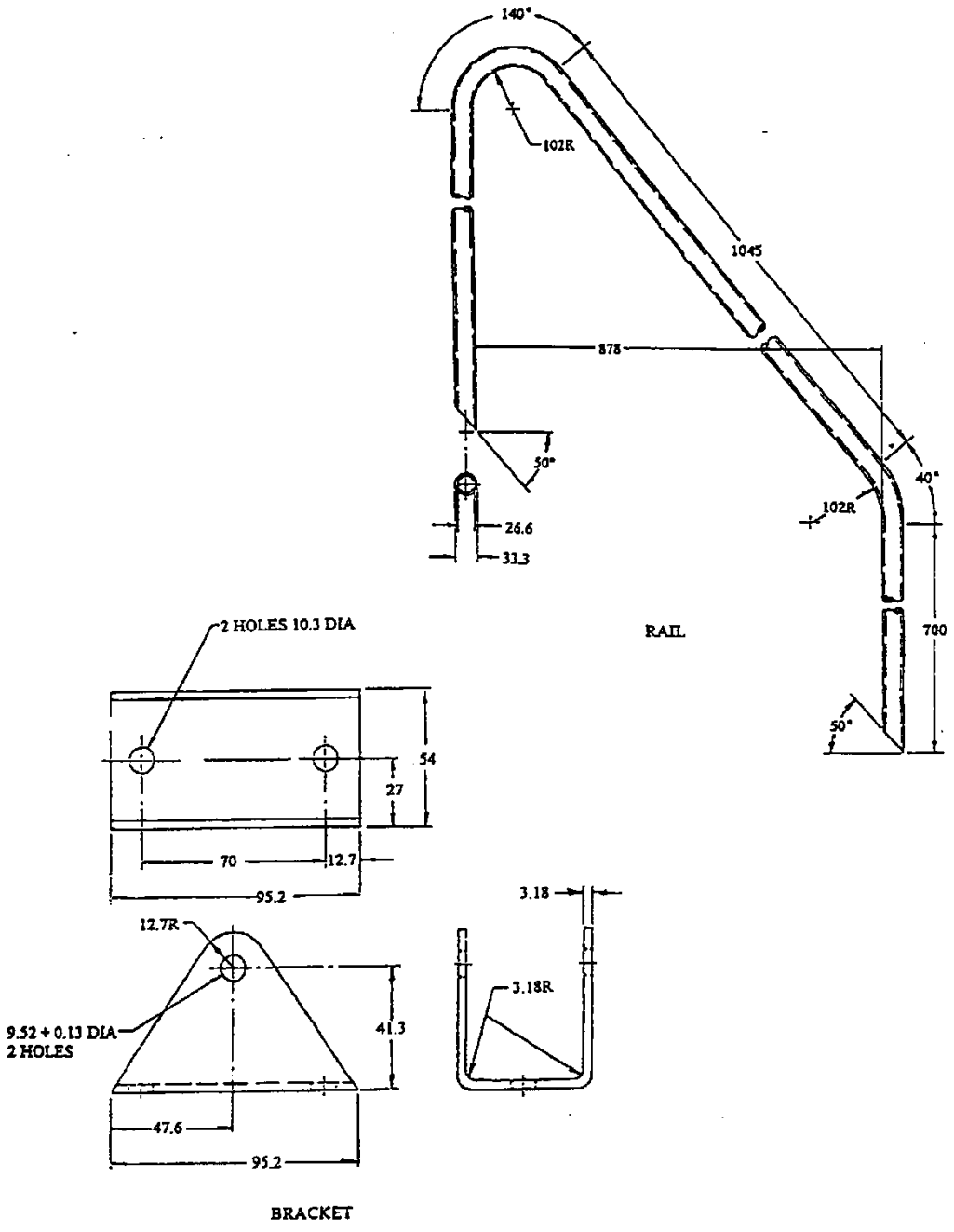
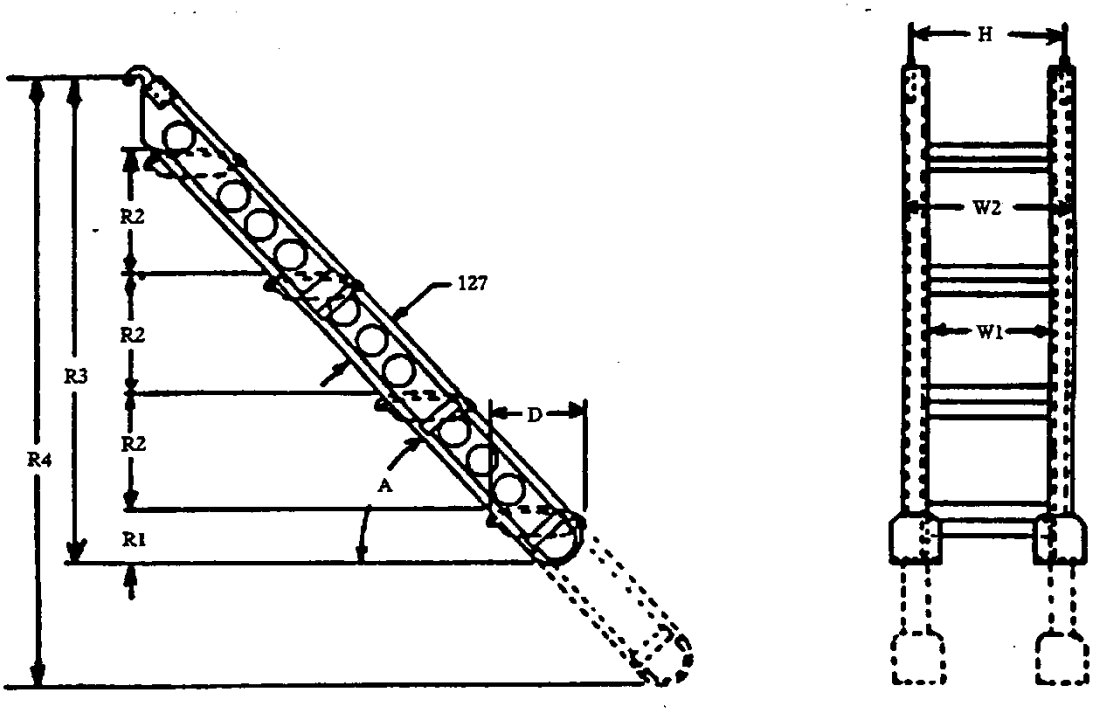



FIGURE 7. Ladder assembly, type G - Continued.



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| | | |
|---|---|---|
| PIN number | | A52163-H1 |
| Number of steps | | 4 |
| Width, inside rails | W1 | 314 mm (12.4 in) |
| Width, outside rails | W2 | 375 mm (14.8 in) |
| Hook centers | H | 367 mm (14.4 in) |
| Step depth | D | 229 mm (9 in) |
| First step rise | R1 | 146 mm (5.75 in) |
| Typical step rise | R2 | 286 mm (11.2 in) |
| Vertical rise, retracted | R3 | 1140 mm (45 in) |
| Vertical rise, extended | R4 | 1430 mm (56.2 in) |
| Angle with ground | A | 48° |
| Material | | Steel |

FIGURE 8. Ladder assembly, type H.



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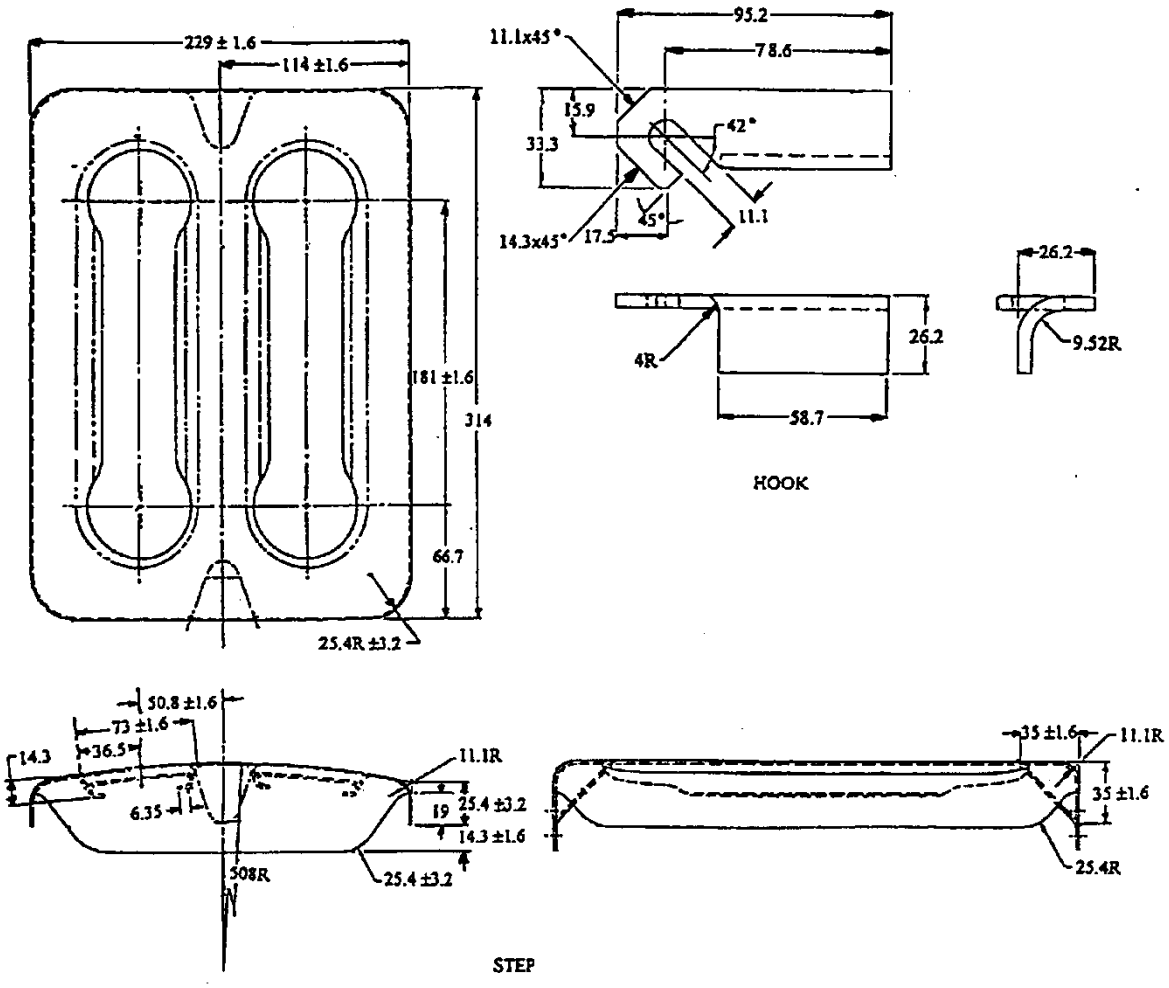
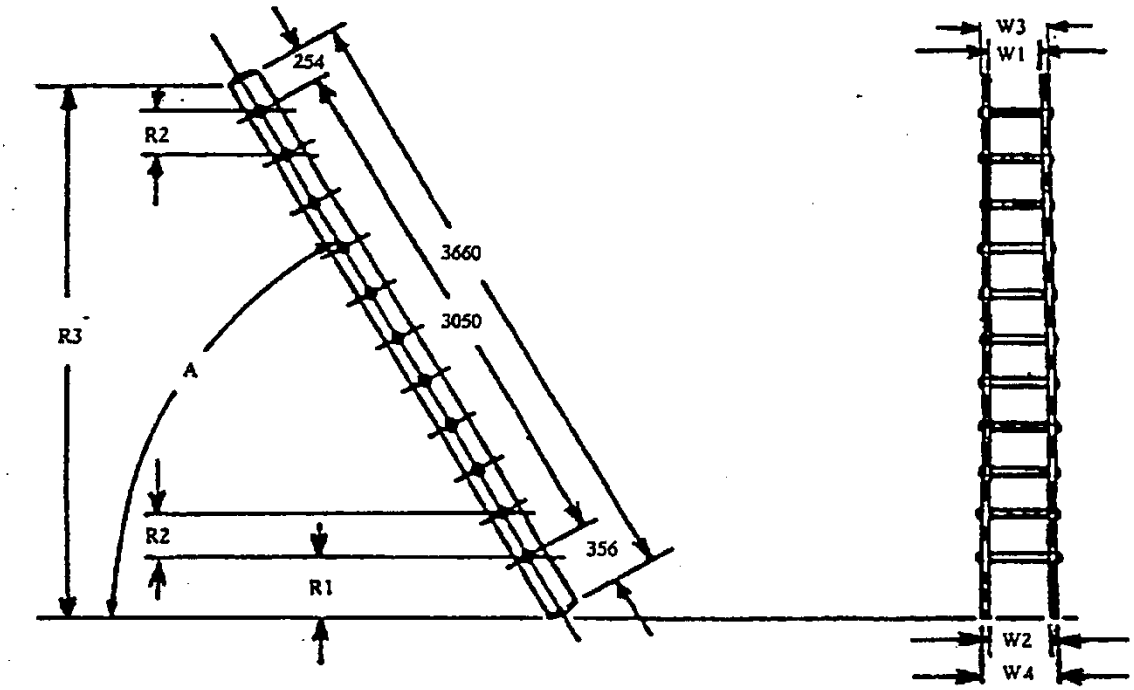



FIGURE 8. Ladder assembly, type H - Continued.



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| | | |
|---|---|---|
| PIN number | | A52163-I1 |
| Number of rungs | | 11 |
| Width, inside top rails | W1 | 292 mm (11.5 in) |
| Width, inside bottom rails | W2 | 368 mm (14.5 in) |
| Width, outside top rails | W3 | 330 mm (13 in) |
| Width, outside bottom rails | W4 | 406 mm (16 in) |
| Angle with ground | A | 60° (approximately) |
| First rung rise | R1 | 308 mm (12.1 in) |
| Typical rung rise | R2 | 264 mm (10.4 in) |
| Vertical rise, total | R3 | 317 mm (12.5 in) |
| Rung section diameters | | 25.4 mm outside diameter (O.D.) x 19.1 mm inside diameter (I.D.) |
| Rail section dimensions | | (1 in O.D. x 0.75 in I.D.) 57.2 x 19.1 x 2.4 mm |
| Material | | (2.25 x 0.75 x 0.094 in) Aluminum |

FIGURE 9. Ladder assembly, type I.



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2.9 Identification markings. The ladder shall be permanently and legibly marked with the following information:

a. "Ladder, vehicle boarding"
b. Manufacturer's identification code (CAGE)
c. Contract number
d. National stock number (NSN)
e. Part identification number (PIN)

3. REGULATORY REQUIREMENTS

3.1 Recovered material. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections, examinations, and tests.

4.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5. PACKAGING. Preservation, packaging, packing, labeling, and marking for the desired level shall be as specified in the contract.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Source of documents.

6.1.1 Government documents. The Code of Federal Regulations (CFR) may be obtained from the www.gpo.gov/fdsys or from Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. Copies of FED-STD-595/34094 may be obtained from



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https://assist.dla.mil/quicksearch/ or the Document Automation and Production Service, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

6.1.2 Government drawings. Army drawings are available from U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397 5000 or can be requested by sending an email to usarmy.detroit.rdecom.mbx.tardec-standardization@mail.mil.

6.1.3 Non-Government publications. Copies of ANSI A14.2, "Portable Metal Ladders" are available from www.ansi.org or the American National Standards Institute (ANSI), 11 West 42nd Street, New York, NY 10036. Copies of ASTM A123/A123M are available from www.astm.org or the American Society for Testing and Materials (ASTM), 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of specifications and standards to be cited in the solicitation.
c. If additional drawings are needed.
d. If CARC paint is required.
e. If camouflage patterns and colors are required.
f. Identification and marking if other than specified.
g. PIN and quantity required.

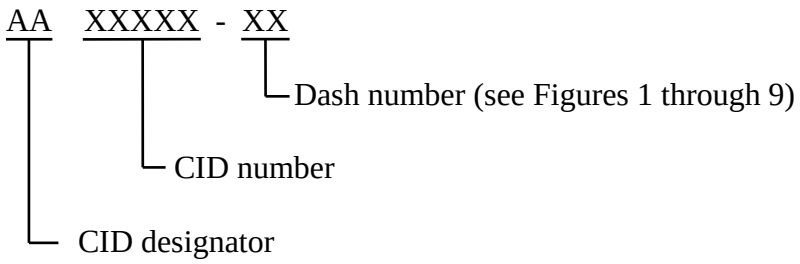


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6.3 Cross-reference data. Ladders conforming to this CID are interchangeable/substitutable with ladders conforming to A-A-52163. Applicable Army drawing numbers cross-referenced to applicable CID PINs are listed as follows:

| Army drawing number | CID PIN number |
|---|---|
| 10882157-1 | A52163-B1 |
| 8389462-1 | A52163-B2 |
| 10882157 | A52163-B3 |
| 8389462 | A52163-B4 |
| 10885309 | A52163-C1 |
| 7034270 | A52163-D1 |
| 8757809 | A52163-D2 |
| 7341560 | A52163-D3 |
| 7341560 | A52163-D4 |
| 7084975 | A52163-E1 |
| 8759434 | A52163-E2 |
| SCD147189 | A52163-F1 |
| SCD108737 | A52163-F2 |
| 8722222 | A52163-G1 |
| 8161317 | A52163-H1 |
| 10891462 | A52163-I1 |

6.4 Part or identification number (PIN). The PIN to be used for ladders acquired to this CID are created as follows:






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MILITARY INTERESTS:

Custodians:
Army - AT
Navy - MC
Air Force - 99

Review Activities:
Army - MI
Air Force - 84

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FAS

Preparing Activity:
Army - AT

(Project 2540-2009-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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